Predictive Stochastic Modeling for Financial Markets

Revised edition

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This is a revised edition of the undergraduate thesis that received the Adam Smith Award. It re-estimates the original models with a corrected data alignment, adds a real-time out-of-sample evaluation, and rewrites the paper around what the evidence supports. The original award version (February 2025) and its R code are preserved unchanged in the repository: github.com/jadenfix/Predictive-Stochastic-Modeling-for-Financial-Markets-.

Abstract

Can market and macroeconomic indicators forecast bear markets in the S&P 500? The award version of this thesis answered yes, but its evidence rested on three timing errors: the Markov-switching model conditioned on next month's price, the "k months ahead" regressions actually explained the regime probability k months in the past, and the out-of-sample "MSPE ratio" divided squared mean errors instead of mean squared errors. I replicate the archived results almost exactly (correlation 1.000 across the reported coefficients), correct each error, and evaluate real-time forecasts made from January 1960 onward with every model re-estimated using only data available at the forecast origin. A two-regime MS-AR(1) model of real returns still separates a calm bull regime from a short, volatile bear regime (expected durations 17.9 and 5.2 months). Out of sample, a logit on 6 state variables reduces the one-month Brier score by 20.8% relative to the unconditional frequency, but simply knowing whether the market is currently in a confirmed bear market achieves 18.9%, and none of the 20 model-horizon pairs beats that persistence benchmark at the 5% level. Beyond three months the predictors add little: at twelve months no model is reliably better than the unconditional frequency. The dividend yield is the most robust single predictor, but its gains are modest. Bear markets are partly identifiable in real time; they are not meaningfully predictable months in advance with these data.

1 Introduction

Bear markets are costly for investors and matter for policy, and whether they can be anticipated is a direct test of how much public information prices already contain. A long literature finds that equity returns alternate between persistent bull and bear regimes (Maheu and McCurdy, 2000; Ang and Timmermann, 2012) and that valuation ratios and macroeconomic conditions carry some information about future returns and downturns (Campbell and Shiller, 1988; Fama and French, 1988; Chen, 2009). The award version of this thesis combined these ideas: it estimated a two-state Markov-switching model on the S&P 500, treated the filtered probability of the bear state as the object to forecast, and related it to lagged market and macroeconomic indicators with OLS and beta regressions. It reported strong in-sample relationships and out-of-sample loss ratios against an intercept-only benchmark as low as 0.004.

This revised edition revisits that analysis with a stricter standard: a forecast may use only information that was published at the time it is made, and forecasting skill is measured only out of sample. Rebuilding the code from scratch revealed three errors in the archived workflow, each of which lets future information or an incorrect loss function flatter the results (Section 7). I first replicate the archived specification including those errors, which confirms that they explain the reported numbers. I then correct them and extend the analysis in three ways. First, I date bear markets with an explicit, reproducible rule so that the target is an observable event rather than a model output. Second, I add logit and probit models, the natural benchmarks for a binary outcome (Estrella and Mishkin, 1998; Chen, 2009). Third, I evaluate real-time forecasts over 1960:02-2023:12 with expanding-window re-estimation and compare them with two simple baselines: the unconditional bear frequency and the persistence of the current, real-time market state.

The corrected evidence is more modest than the original. Regime-switching structure in returns is strong and stable. Indicators such as inflation, volatility and the dividend yield are associated with bear markets in sample. In real time, however, most of the short-horizon forecastability comes from recognizing a bear market that is already under way, and little survives at horizons of six months or more.

2 Related literature

Regime-switching models in the tradition of Hamilton (1989) describe the stock market as moving between a high-return, low-volatility state and a low-return, high-volatility state. Maheu and McCurdy (2000) identify bull and bear markets this way, and Ang and Timmermann (2012) survey the evidence that regime changes are a pervasive feature of financial markets. Rule-based dating is the main alternative: Pagan and Sossounov (2003) adapt business-cycle turning-point algorithms to equity prices, and Lunde and Timmermann (2004) define bull and bear markets by threshold movements from the latest peak or trough. Chen (2009) and Chen, Chen and Chou (2017) use such dates as the dependent variable in probit models and find that macroeconomic and credit variables help predict bear markets.

On the predictor side, Campbell and Shiller (1988) and Fama and French (1988) document that dividend yields forecast long-horizon returns, while Goetzmann and Jorion (1993), Stambaugh (1999) and Lewellen (1999) show that inference with persistent, price-scaled predictors is delicate. Pontiff and Schall (1998) report related evidence for book-to-market ratios, and Hammerschmidt and Lohre (2018) find that return predictability itself varies across regimes. Welch and Goyal (2008) emphasize that many predictors that look strong in sample fail out of sample, which is the standard this edition adopts.

3 Data

3.1 Source and timing

The data are the monthly U.S. stock market series compiled by Robert Shiller (Shiller, 2015), archived in the repository as `one45.csv`. The file covers January 1900 to March 2024 I use observations through December 2023, the last month with complete dividend and earnings data. One trailing row of the file is a text footnote and is dropped. The S&P Composite price is a monthly average of daily closes, so it is known at the end of each month. I treat the end of month t as the forecast origin and impose publication lags on everything else: the consumer price index enters with a lag of 1 month, and dividends and earnings, which Shiller interpolates from quarterly reports, enter with a lag of 6 months. Real values are obtained by deflating with the CPI available at t. Table 1 lists the predictors, their construction and their publication lags.

The archived file also contains columns labeled `realpricelag` to `realpricelag4`. An audit of the file shows that these are leads, not lags: on 1,490 of 1,490 comparable rows, `realpricelag` equals the following month's real price, and it matches the previous month's price on only 3 rows. The corrected pipeline never reads these columns, and an automated test confirms that every predictor is unchanged when all data after its forecast origin are altered.

3.2 Dating bear markets

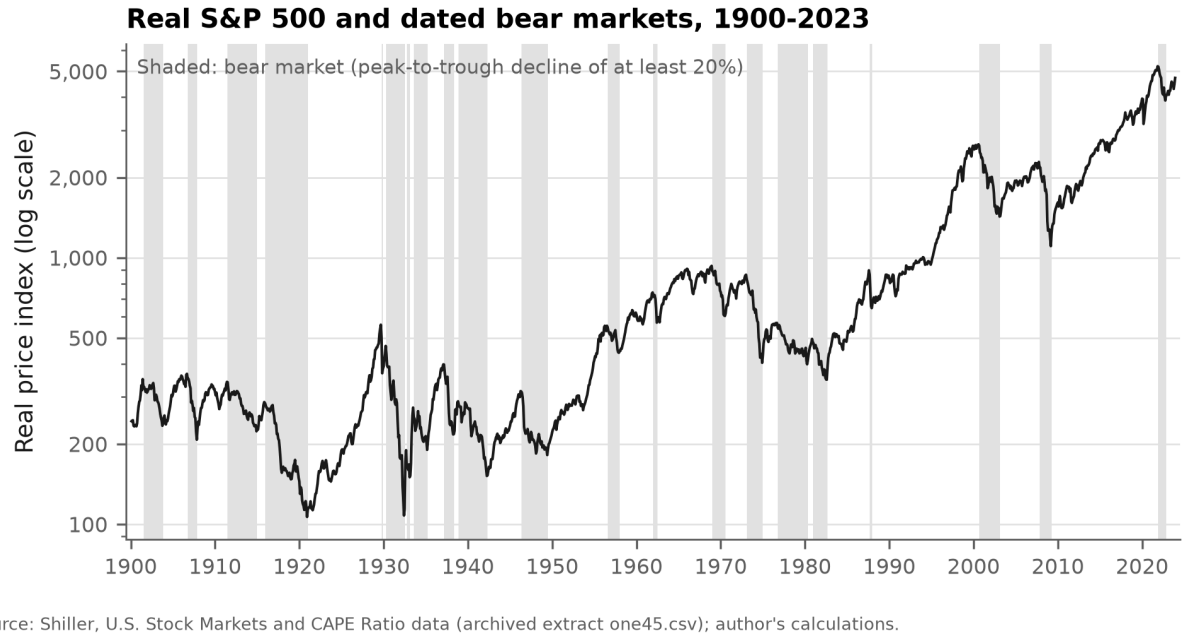
I date bear markets on the real price index with a symmetric threshold rule in the spirit of Lunde and Timmermann (2004). Starting from a bull state, the running maximum is a candidate peak; when the index falls 20% below it, the peak is confirmed and a bear market begins in the following month. In the bear state the running minimum is a candidate trough; when the index rises 20% above it, the trough is confirmed and the bear market ends at the trough month. The rule yields 21 bear markets since 1900, 9 of them beginning after 1960 (Table 2, Figure 1). The median bear market lasts 20 months with a median peak-to-trough decline of 34.8%. Bear months account for 32.4% of the full sample and 22.7% of January 1960 to December 2023.

Table 1: Predictors, transformations and publication lags

Predictor	Construction	Units	Lag (months)	Mean	SD
Monthly-change predictors (archived thesis variables)					
CPI inflation, monthly	100 dlog CPI	% per month	1	0.31	0.36
Change in 10-year yield	Monthly change	pp	0	0.00	0.27
Real bond return	100 dlog real bond index	% per month	0 (CPI 1)	0.18	1.99
Real dividend growth	100 dlog real dividends	% per month	6	0.17	0.63
Real total return	Price return plus D/12P, less inflation	% per month	0 (D 6, CPI 1)	0.49	3.65
TR-scaled earnings growth	100 dlog real TR-scaled earnings	% per month	6	0.45	4.84
Change in log TR-CAPE	100 dlog TR price / 10-yr mean earnings	% per month	0 (E 6)	0.05	3.66
State predictors (bear-market forecasting models)					
Log dividend yield	log(D / P)	log points	0 (D 6)	−3.67	0.40
Log CAPE	log(P / 10-yr mean real earnings)	log points	0 (E 6)	2.98	0.43
CPI inflation, 12-month	100 x 12-month dlog CPI	%	1	3.66	2.68
12-month change in 10-year yield	12-month change	pp	0	−0.01	1.13
12-month real price change	100 x 12-month dlog real price	%	0 (CPI 1)	3.07	15.94
12-month realized volatility	sd of monthly real returns x sqrt(12)	% annualized	0 (CPI 1)	11.25	4.93

Notes: Statistics over 1960:01-2023:12. A forecast is made at the end of month t. Lag gives the publication lag applied to the inputs: CPI enters with one month, dividends (D) and earnings (E) with six. Real values are deflated by the CPI available at t. The archived 'realpricelag' columns (future prices) are never used.

The dated labels are ex post: a peak is only known to have been a peak once the 20% decline has occurred, which took a median of 9 months. The same rule run on data through month t gives the real-time state, which is observable at t, and "vintage" labels for all earlier months. The out-of-sample exercise trains every model on vintage labels only.



Source: Shiller, U.S. Stock Markets and CAPE Ratio data (archived extract one45.csv); author's calculations.

Figure 1: Real S&P 500 price index (log scale) with dated bear markets shaded.

Table 2: Dated bear markets in the real S&P 500, 1900-2023

Peak	Trough	Months	Decline (%)	Confirmed	Months to confirm
1901-06	1903-10	28	−33.2	1903-06	24
1906-09	1907-11	14	−43.4	1907-03	6
1911-06	1914-12	42	−34.8	1913-05	23
1915-11	1920-12	61	−62.9	1917-05	18
1929-09	1929-11	2	−34.2	1929-11	2
1930-04	1932-06	26	−76.9	1930-10	6
1932-09	1933-02	5	−20.8	1933-02	5
1933-07	1935-03	20	−30.6	1934-07	12
1937-02	1938-04	14	−45.4	1937-09	7
1938-11	1942-04	41	−47.5	1940-06	19
1946-04	1949-06	38	−42.4	1946-09	5
1956-07	1957-12	17	−20.8	1957-11	16
1961-12	1962-06	6	−23.0	1962-06	6
1968-12	1970-07	19	−35.1	1970-01	13
1973-01	1974-12	23	−53.3	1973-12	11
1976-09	1980-04	43	−30.0	1978-01	16
1980-11	1982-08	21	−29.7	1982-01	14
1987-08	1987-12	4	−27.9	1987-11	3
2000-08	2003-02	30	−46.4	2001-03	7
2007-10	2009-03	17	−51.7	2008-07	9
2021-11	2022-10	11	−25.6	2022-06	7

Notes: A bear market runs from a peak to the following trough when the real index falls at least 20% below the peak before rising 20% above the trough. 'Confirmed' is the first month in which the 20% decline had occurred, that is, when the bear market became knowable in real time.

4 Methodology

4.1 Markov-switching model of returns

Let $r_t = 100 \cdot \Delta \ln P_t$ be the monthly real return. The main regime model is a two-state Markov-switching autoregression with switching mean and variance (Hamilton, 1989):

$$r_t - \mu_{S_t} = \varphi\big(r_{t-1} - \mu_{S_{t-1}}\big) + \sigma_{S_t}\varepsilon_t, \quad \varepsilon_t \sim N(0,1), \quad S_t \in \{\text{bull}, \text{bear}\}, \tag{1}$$

where S_t follows a first-order Markov chain with transition probabilities $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$. Setting $\varphi = 0$ gives the switching mean-variance model. The bear regime is the one with the lower mean. The expected duration of regime i is $1/(1 - p_{ii})$. The Hamilton filter delivers the filtered probability

$$Q_t = \Pr(S_t = \text{bear} \mid r_1, ..., r_t), \tag{2}$$

which uses only returns through t and is therefore a legitimate real-time quantity once the parameters are fixed. The smoothed probability $\Pr(S_t = \text{bear} \mid r_1, ..., r_T)$ conditions on the whole sample, so I use it only for retrospective description. An h-step regime forecast is $\Pr(S_{t+h} = \text{bear} \mid r_1, ..., r_t) = [Q_t^c, Q_t]\boldsymbol{P}^h e_2$, where $\boldsymbol{P}$ is the transition matrix and $Q_t^c = 1 - Q_t$.

4.2 Predictive regressions for the regime probability

Following the original thesis, each predictor x_t enters a separate regression for the regime probability h months ahead:

$$Q_{t+h} = \alpha + \beta x_t + e_{t+h}. \tag{3}$$

I estimate Equation 3 by OLS with Newey and West (1987) standard errors, since overlapping horizons and the persistence of Q_t make the errors serially correlated. Because Q_t lies in the unit interval, I also estimate a beta regression (Ferrari and Cribari-Neto, 2004),

$$Q_{t+h} \sim \text{Beta}(\mu_t\varphi, (1-\mu_t)\varphi), \quad \text{logit}(\mu_t) = \alpha + \beta x_t, \tag{4}$$

after the standard compression of Q into the open interval (Smithson and Verkuilen, 2006), again with Newey-West standard errors. Predictors are standardized, so β measures the effect of a one-standard-deviation change. The test of $\beta = 0$ is the test of no predictability that the original thesis framed in terms of market efficiency.

4.3 Binary models for dated bear markets

The regime probability is itself an estimate. The dated bear indicator y_{t+h} is an observable target, and the natural model for it is a binary regression,

$$\Pr(y_{t+h} = 1 \mid x_t) = F(\alpha + x_t^\top \beta), \tag{5}$$

with F the logistic (logit) or standard normal (probit) distribution function. The 6 predictors in x_t are the log dividend yield, the log CAPE ratio with lagged earnings, twelve-month CPI inflation, the twelve-month change in the ten-year Treasury yield, the twelve-month real price change, and twelve-month realized volatility.

4.4 Out-of-sample design

Forecasts are made at every month-end from January 1960 onward for horizons of 1, 3, 6, 12 months. At each origin t every model is re-estimated on an expanding window using only (i) predictors dated $s \leq t$, (ii) vintage bear labels from dating the price history through t, so that a training pair (x_s, y_{s+h}) requires $s + h \leq t$, and (iii) for the regime model, parameters re-estimated each January on returns through the origin (64 estimations) and filtered probabilities from those parameters. The training data start in June 1910, when all predictors are first available. I compare 7 forecasts:

- **Unconditional frequency**: the training-sample share of bear months h months ahead.
- **Persistence**: the training-sample frequency of a bear market h months ahead, conditional on the current real-time state (in a confirmed bear market or not).
- **MS-AR(1)**: a logit recalibration of the h-step regime forecast from the filtered probability.
- **Logit: dividend yield**, the single predictor emphasized by the original thesis.
- **Logit** and **probit** on all 6 predictors.
- **Logit: all predictors + state**, which adds the real-time state to test whether the predictors add to persistence.

Accuracy is measured by the Brier score (Brier, 1950), the log loss (with probabilities truncated to $[0.001, 1 - 0.001]$), the area under the ROC curve, and calibration curves. Brier skill is the percentage reduction in the Brier score relative to the unconditional frequency. Differences in Brier score are tested with the Diebold and Mariano (1995) statistic with the small-sample correction of Harvey, Leybourne and Newbold (1997). For forecasts of the regime probability itself (Equation 3 and Equation 4), I report ratios of mean squared prediction errors against the intercept-only model and the Clark and West (2007) test for nested models.

5 In-sample results

5.1 Regimes in real returns

Table 3 reports the regime models. The structure is clear and stable across specifications and samples. In the MS-AR(1) model for January 1960 to December 2023, the bull regime has a mean real return of 0.84% per month with a standard deviation of 2.45%, while the bear regime has a mean of −1.81% and a standard deviation of 5.62%. Both regimes are persistent ($p_{\text{bull,bull}} = 0.944$, $p_{\text{bear,bear}} = 0.808$), implying expected durations of 17.9 months for bull regimes and 5.2

Table 3: Two-regime Markov-switching models of monthly real S&P 500 returns

	MS, 1960-2023	MS-AR(1), 1960-2023	MS, 1900-2023	MS-AR(1), 1900-2023
Mean monthly real return (%)				
Bull regime	0.900^{***}	0.835^{***}	0.773^{***}	0.711^{***}
	(0.131)	(0.144)	(0.097)	(0.117)
Bear regime	-1.887^{***}	-1.812^{***}	-2.221^{***}	-2.231^{***}
	(0.585)	(0.668)	(0.535)	(0.651)
Innovation variance				
Bull regime	6.35	6.01	8.24	7.88
	(0.55)	(0.55)	(0.48)	(0.48)
Bear regime	30.94	31.54	56.48	55.64
	(4.29)	(4.51)	(6.17)	(6.33)
Bull volatility (sd, %)	2.52	2.45	2.87	2.81
Bear volatility (sd, %)	5.56	5.62	7.52	7.46
Autoregression				
AR(1) coefficient		0.194^{***}		0.232^{***}
		(0.035)		(0.026)
Transitions				
P(bull to bull)	0.949	0.944	0.967	0.967
	(0.017)	(0.019)	(0.008)	(0.009)
P(bear to bear)	0.826	0.808	0.860	0.855
	(0.055)	(0.065)	(0.035)	(0.039)
Expected bull duration (months)	19.5	17.9	30.5	30.4
Expected bear duration (months)	5.7	5.2	7.2	6.9
Fit				
Log-likelihood	−2,007.0	−1,990.3	−4,069.3	−4,028.8
AIC	4,026.1	3,994.5	8,150.5	8,071.5
BIC	4,053.9	4,027.0	8,182.4	8,108.7
Observations	768	767	1,486	1,485
Sample	1960:01-2023:12	1960:02-2023:12	1900:03-2023:12	1900:04-2023:12

Notes: Maximum-likelihood estimates; standard errors in parentheses from the numerical Hessian. The bear regime is the regime with the lower mean. Switching mean and variance; the AR(1) coefficient is common to both regimes. Returns are 100 times the log change in the real price index. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided). Stars are shown only for tests against zero that are meaningful (means, AR term).

months for bear regimes, close to the estimates in the literature that the original thesis cited. The autoregressive term ($\varphi = 0.194$) partly reflects the time averaging of Shiller's monthly prices and lowers the AIC from 4,026.1 to 3,994.5, so the MS-AR(1) is the main specification. On the longer 1900 to 2023 sample the bull regime is more persistent (expected duration 30.4 months) and the bear regime is more volatile.

Figure 2 contrasts the filtered and smoothed bear-regime probabilities. The smoothed series marks the dated bear markets more cleanly (AUC 0.859 against 0.799 for the contemporaneous dated state), but that advantage comes from conditioning on later returns. The filtered series is noisier, reacting to individual volatile months such as October 1987 and early 2020, and it is the only one a forecaster could have used.

5.2 The corrected archived specification

Table 4 re-estimates the original regressions of the regime probability on the 7 original predictors with the timing corrected: the target is Q_{t+h}, the predictors are observable at t, and index levels are replaced by growth rates so that coefficients are interpretable. The number of predictors significant at the 5% level in the OLS regressions is 4 of 7 at one month, 3 at three, 6 at six and 2 at twelve. The largest R^2 (0.121) belongs to the predictor "Real total return" at one month: a one-standard-deviation higher real return lowers next month's bear probability by -9.20^{***} (1.12) percentage

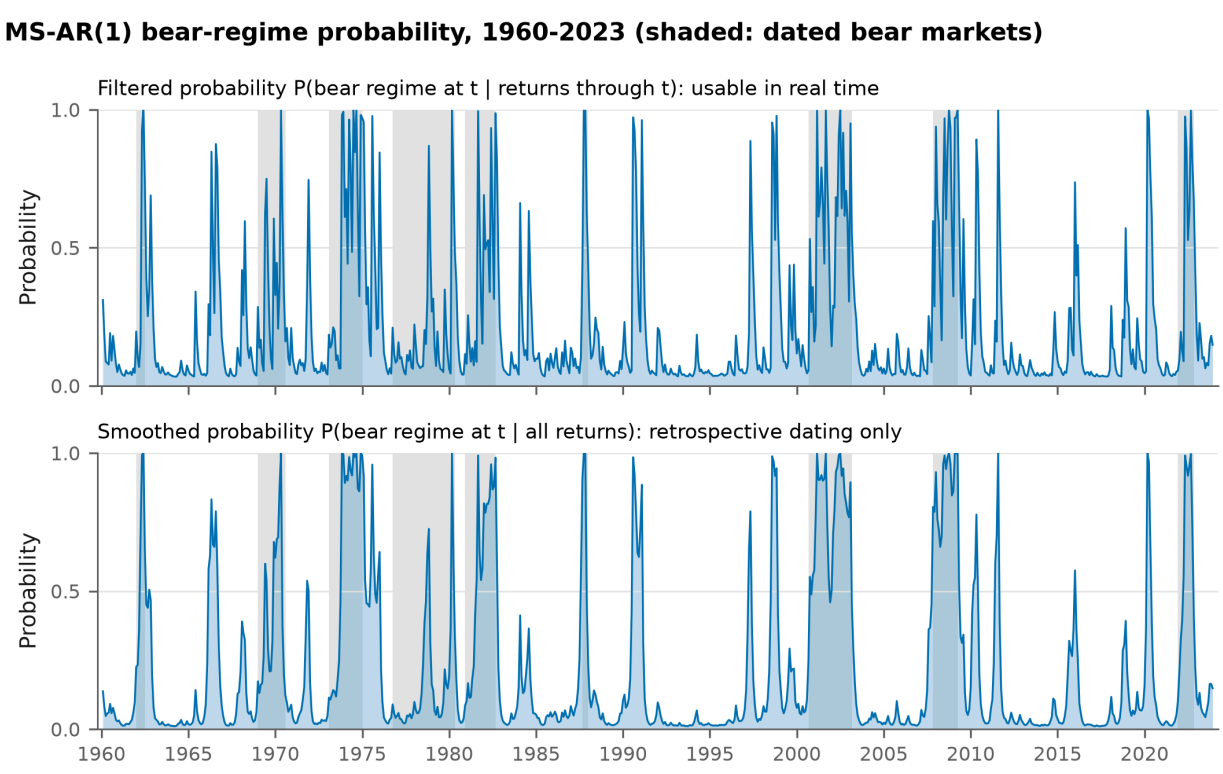


Figure 2: Filtered and smoothed bear-regime probabilities from the MS-AR(1) model, with dated bear markets shaded. Only the filtered probability is used for forecasting.

points. This is close to mechanical, because the filter itself updates on recent returns. Dividend growth (-3.80^{***} (1.33) at one month) and inflation have the expected signs. Total-return-scaled earnings, which the award version singled out, show no meaningful relationship once the timing is corrected ($R^2 = 0.001$ at one month).

5.3 Logit and probit models of dated bear markets

Table 5 reports full-sample binary models for the dated bear indicator, and Figure 3 plots the logit coefficients. Higher inflation (0.82^{***} (0.26) at one month) and higher realized volatility (0.51^{***} (0.18) at six months) raise the probability of a bear market at all horizons. A strong past year lowers the short-horizon probability (-0.78^{***} (0.25) at one month), which captures persistence: a low twelve-month return usually means a decline is already under way. The dividend yield enters positively (0.97^{**} (0.44) at six months). Because it divides lagged dividends by the current price, it rises when prices fall, so part of this coefficient also reflects an ongoing decline rather than cheapness. The dividend yield and the CAPE ratio are strongly negatively correlated, so their conditional coefficients should not be read separately. The pseudo-R^2 falls from 0.261 at one month to 0.105 at twelve months.

Table 4: Corrected archived specification: regime probability h months ahead on each predictor

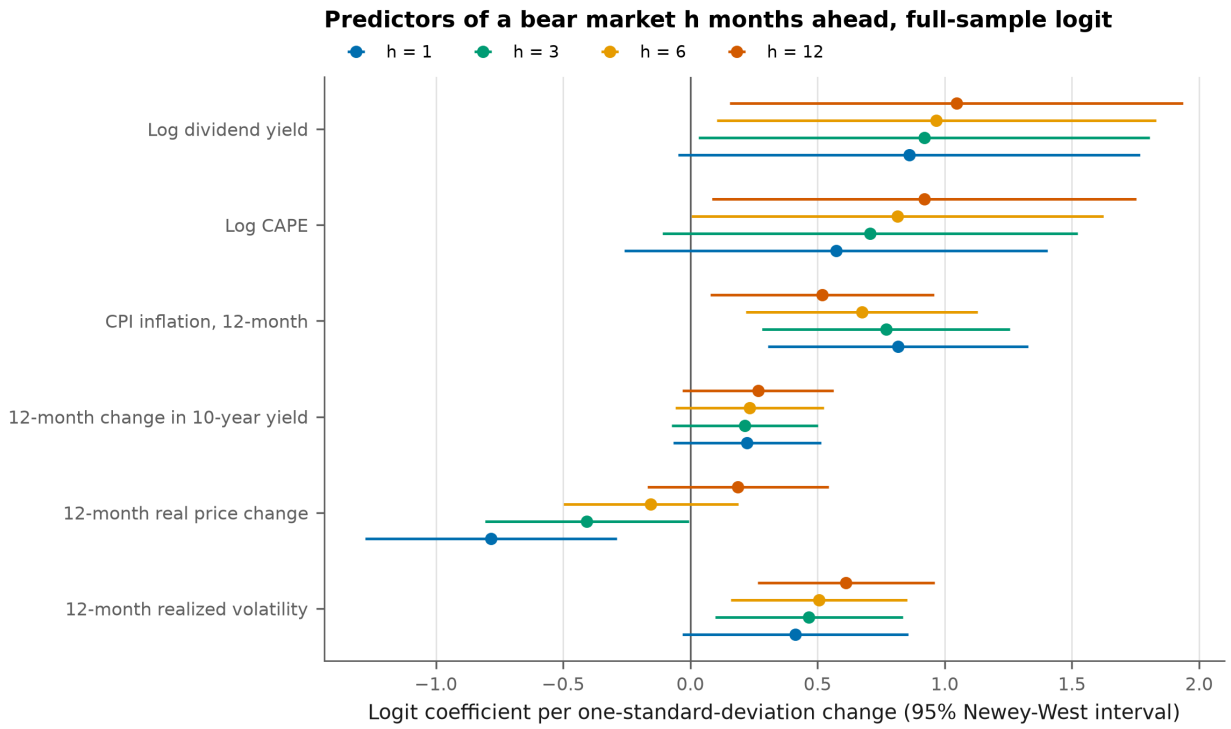
	OLS (percentage points)				Beta regression (logit link)			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
CPI inflation, monthly	3.643**	4.065*	4.512**	4.267***	0.155**	0.200**	0.180**	0.171***
	(1.824)	(2.282)	(1.760)	(1.530)	(0.076)	(0.095)	(0.071)	(0.061)
R^2 / pseudo-R^2	0.019	0.024	0.029	0.026	0.020	0.027	0.029	0.030
Change in 10-year yield	−0.066	−1.349	3.575***	0.944	0.017	−0.052	0.131***	0.036
	(1.329)	(1.100)	(1.026)	(0.815)	(0.060)	(0.040)	(0.043)	(0.028)
R^2 / pseudo-R^2	0.000	0.003	0.018	0.001	0.000	0.003	0.017	0.001
Real bond return	−0.104	1.060	−3.911***	−1.386	−0.018	0.039	−0.145***	−0.052*
	(1.412)	(1.303)	(1.212)	(0.909)	(0.060)	(0.048)	(0.049)	(0.031)
R^2 / pseudo-R^2	0.000	0.002	0.022	0.003	0.000	0.001	0.021	0.003
Real dividend growth	−3.805***	−3.337**	−3.395**	−2.808	−0.122***	−0.112**	−0.115**	−0.111
	(1.329)	(1.344)	(1.388)	(2.034)	(0.047)	(0.044)	(0.047)	(0.073)
R^2 / pseudo-R^2	0.021	0.016	0.017	0.011	0.017	0.014	0.015	0.015
Real total return	−9.195***	−6.111***	−5.763***	−1.306	−0.411***	−0.241***	−0.218***	−0.039
	(1.118)	(1.099)	(1.295)	(0.981)	(0.045)	(0.047)	(0.052)	(0.032)
R^2 / pseudo-R^2	0.121	0.053	0.048	0.002	0.107	0.049	0.043	0.002
TR-scaled earnings growth	−0.859	−1.595	−2.083	−2.630**	−0.082	−0.065	−0.066	−0.085*
	(2.498)	(1.782)	(1.660)	(1.309)	(0.123)	(0.073)	(0.062)	(0.049)
R^2 / pseudo-R^2	0.001	0.004	0.006	0.010	0.003	0.003	0.005	0.007
Change in log TR-CAPE	−9.128***	−6.109***	−5.849***	−1.507	−0.407***	−0.242***	−0.222***	−0.046
	(1.115)	(1.116)	(1.289)	(1.011)	(0.044)	(0.047)	(0.052)	(0.033)
R^2 / pseudo-R^2	0.120	0.053	0.049	0.003	0.106	0.049	0.045	0.002
Observations	766	764	761	755	766	764	761	755

Notes: Each cell is a separate univariate regression of the filtered bear-regime probability Q(t+h) from the MS-AR(1) model on the predictor at t, which is standardized (coefficients are per one standard deviation). OLS coefficients are in percentage points of probability. Newey-West standard errors in parentheses (7, 12 lags by horizon). Forecast origins 1960:02 to 2023:11 for h = 1; the sample shortens by h months. The beta-regression pseudo-R^2 is the squared correlation of the linear predictor with logit(Q). * p < 0.10, ** p < 0.05, *** p < 0.01 (two-sided). Q is computed with full-sample parameters, so these are in-sample associations, not forecasts.

Table 5: Full-sample logit and probit models of a dated bear market h months ahead

	Logit				Probit			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
Valuation								
Log dividend yield	0.86*	0.92**	0.97**	1.05**	0.49*	0.53**	0.57**	0.62**
	(0.46)	(0.45)	(0.44)	(0.46)	(0.25)	(0.25)	(0.25)	(0.26)
Log CAPE	0.57	0.71*	0.81**	0.92**	0.32	0.40*	0.48**	0.55**
	(0.42)	(0.42)	(0.41)	(0.43)	(0.23)	(0.24)	(0.24)	(0.25)
Macroeconomic								
CPI inflation, 12-month	0.82***	0.77***	0.67***	0.52**	0.46***	0.43***	0.39***	0.31**
	(0.26)	(0.25)	(0.23)	(0.22)	(0.14)	(0.14)	(0.13)	(0.13)
12-month change in 10-year yield	0.22	0.21	0.23	0.27*	0.14*	0.14	0.14	0.15*
	(0.15)	(0.15)	(0.15)	(0.15)	(0.09)	(0.09)	(0.09)	(0.09)
Market								
12-month real price change	−0.78***	−0.41**	−0.16	0.19	−0.43***	−0.24**	−0.10	0.10
	(0.25)	(0.20)	(0.18)	(0.18)	(0.12)	(0.11)	(0.10)	(0.11)
12-month realized volatility	0.41*	0.47**	0.51***	0.61***	0.25**	0.28***	0.30***	0.37***
	(0.23)	(0.19)	(0.18)	(0.18)	(0.11)	(0.10)	(0.10)	(0.10)
Constant	−0.95***	−0.89***	−0.85***	−0.82***	−0.58***	−0.54***	−0.52***	−0.49***
	(0.18)	(0.17)	(0.18)	(0.19)	(0.10)	(0.10)	(0.10)	(0.11)
Pseudo-R^2 (McFadden)	0.261	0.186	0.137	0.105	0.260	0.186	0.137	0.104
Log-likelihood	−633.3	−697.1	−737.6	−763.0	−634.2	−696.9	−737.4	−763.6
AIC	1,280.7	1,408.3	1,489.1	1,540.1	1,282.4	1,407.9	1,488.8	1,541.2
Observations	1,362	1,360	1,357	1,351	1,362	1,360	1,357	1,351

Notes: Dependent variable: 1 if month t+h lies in a dated bear market. Predictors are measured at t and standardized. Newey-West standard errors in parentheses (9, 9, 9, 12 lags by horizon). Forecast origins 1910:06 to 2023:11 for h = 1. * p < 0.10, ** p < 0.05, *** p < 0.01 (two-sided). In-sample estimates over the whole period, not real-time forecasts.



Source: Shiller, U.S. Stock Markets and CAPE Ratio data (archived extract one45.csv); author's calculations.

Figure 3: Full-sample logit coefficients per one-standard-deviation change in each predictor, with 95% Newey-West confidence intervals, by forecast horizon.

6 Out-of-sample evaluation

6.1 Forecasting dated bear markets

Table 6 is the central result of this edition. At the one-month horizon every model beats the unconditional frequency (Brier 0.198): the multivariate logit reaches 0.157 (skill 20.8%, AUC 0.780) and the probit 0.155. But persistence alone reaches 0.161 (skill 18.9%), and the Diebold-Mariano tests cannot distinguish the multivariate models from persistence ($p = 0.316$ for the logit and 0.256 for the probit). Adding the real-time state to the logit does not help (0.159). The MS-AR(1) forecast is informative (AUC 0.702) but less accurate than persistence (skill 10.9%).

At three months the single-variable dividend-yield logit is the most accurate model (Brier 0.175, skill 12.3%), essentially tied with persistence (skill 11.5%, $p = 0.433$), while the multivariate logit falls to a skill of 6.2%. At six months only the dividend-yield logit retains a meaningful gain (skill 7.9%, $p = 0.051$ against the unconditional frequency), and the multivariate models are worse than the unconditional frequency (skill −6.9%). At twelve months no model is reliably better than the unconditional frequency: the best skill is 1.9% ($p = 0.353$), and the multivariate logit has an AUC of 0.417, below the 0.5 of an uninformative forecast. Across all 20 comparisons of a model with persistence, 0 are significant at the 5% level (smallest $p = 0.256$).

Figure 4 shows why. The model probabilities rise sharply once a decline is under way and fall when the market recovers, so at short horizons they track bear markets well, mostly with a lag. In the months before the peaks of 1973, 2000 and 2007 the model probabilities were below the unconditional frequency. The calibration curves in Figure 5 show that the short-horizon forecasts are directionally right but overconfident at the top: months assigned the highest probabilities

Table 6: Out-of-sample accuracy of real-time bear-market forecasts, targets 1960:02-2023:12

	Brier	Skill (%)	Log loss	AUC	p vs uncond.	p vs persist.
h = 1 (767 forecasts, bear share 22.7%)						
Unconditional frequency	0.198	0.0	0.587	0.563		
Persistence (real-time state)	0.161	18.9	0.502	0.688	<0.001	
MS-AR(1) filtered probability	0.176	10.9	0.537	0.702	<0.001	0.987
Logit: dividend yield	0.165	16.5	0.527	0.627	<0.001	0.680
Logit: all predictors	0.157	20.8	0.488	0.780	<0.001	0.316
Probit: all predictors	0.155	21.5	0.487	0.779	<0.001	0.256
Logit: all predictors + state	0.159	19.9	0.497	0.772	<0.001	0.387
h = 3 (765 forecasts, bear share 22.7%)						
Unconditional frequency	0.199	0.0	0.589	0.547		
Persistence (real-time state)	0.176	11.5	0.537	0.636	<0.001	
MS-AR(1) filtered probability	0.193	3.0	0.575	0.618	0.182	0.994
Logit: dividend yield	0.175	12.3	0.545	0.586	0.001	0.433
Logit: all predictors	0.187	6.2	0.563	0.675	0.134	0.878
Probit: all predictors	0.185	7.0	0.561	0.676	0.099	0.838
Logit: all predictors + state	0.189	5.1	0.573	0.667	0.197	0.933
h = 6 (762 forecasts, bear share 22.8%)						
Unconditional frequency	0.200	0.0	0.592	0.528		
Persistence (real-time state)	0.190	4.9	0.568	0.571	0.094	
MS-AR(1) filtered probability	0.205	−2.5	0.602	0.530	0.737	0.986
Logit: dividend yield	0.185	7.9	0.567	0.541	0.051	0.287
Logit: all predictors	0.214	−6.9	0.623	0.549	0.843	0.981
Probit: all predictors	0.213	−6.3	0.621	0.548	0.826	0.977
Logit: all predictors + state	0.218	−8.6	0.634	0.537	0.879	0.992
h = 12 (756 forecasts, bear share 23.0%)						
Unconditional frequency	0.202	0.0	0.595	0.513		
Persistence (real-time state)	0.201	0.8	0.591	0.513	0.331	
MS-AR(1) filtered probability	0.209	−3.5	0.609	0.463	0.847	0.959
Logit: dividend yield	0.198	1.9	0.594	0.488	0.353	0.405
Logit: all predictors	0.224	−11.1	0.654	0.417	0.911	0.939
Probit: all predictors	0.226	−11.8	0.659	0.411	0.924	0.949
Logit: all predictors + state	0.227	−12.3	0.660	0.407	0.929	0.952

Notes: Expanding-window forecasts made at each month-end origin using only data available then: predictors with publication lags, bear dates obtained from prices through the origin, and Markov-switching parameters re-estimated each January (filtered probabilities only). Lower Brier score and log loss are better; skill is the percentage reduction in Brier score relative to the unconditional frequency. Log loss truncates probabilities to [0.001, 0.999]. The p-values are one-sided Diebold-Mariano tests (Harvey-Leybourne-Newbold correction, Bartlett HAC variance) of equal Brier score against the alternative that the model is more accurate than the baseline.

turned out to be bear months less often than predicted. At twelve months the curves are flat, and the ROC curves in Figure 6 lie close to the diagonal.

6.2 Forecasting the regime probability

Table 7 revisits the original out-of-sample exercise, in which each predictor forecasts the regime probability and is compared with an intercept-only model. Panel A reproduces the archived numbers with the archived code path and formula: 22 of 28 OLS ratios fall below one. Panel B keeps the archived design but computes true mean squared errors; only 11 do. Panel C uses the corrected design with expanding-window refits from April 2011. Here 16 of 28 OLS ratios fall below one and 13 are significant at the 10% level by the Clark-West test. The gains are real but small: the best ratio is 0.865, for the predictor "Change in log TR-CAPE" at 1 month. For comparison, the regime probability's own current value achieves a ratio of 0.476 at one month, but none at six months (1.032), where inflation and dividend growth still carry a few percent of information. The intercept-only model is a weak benchmark for a target as persistent as Q_t, and this panel uses

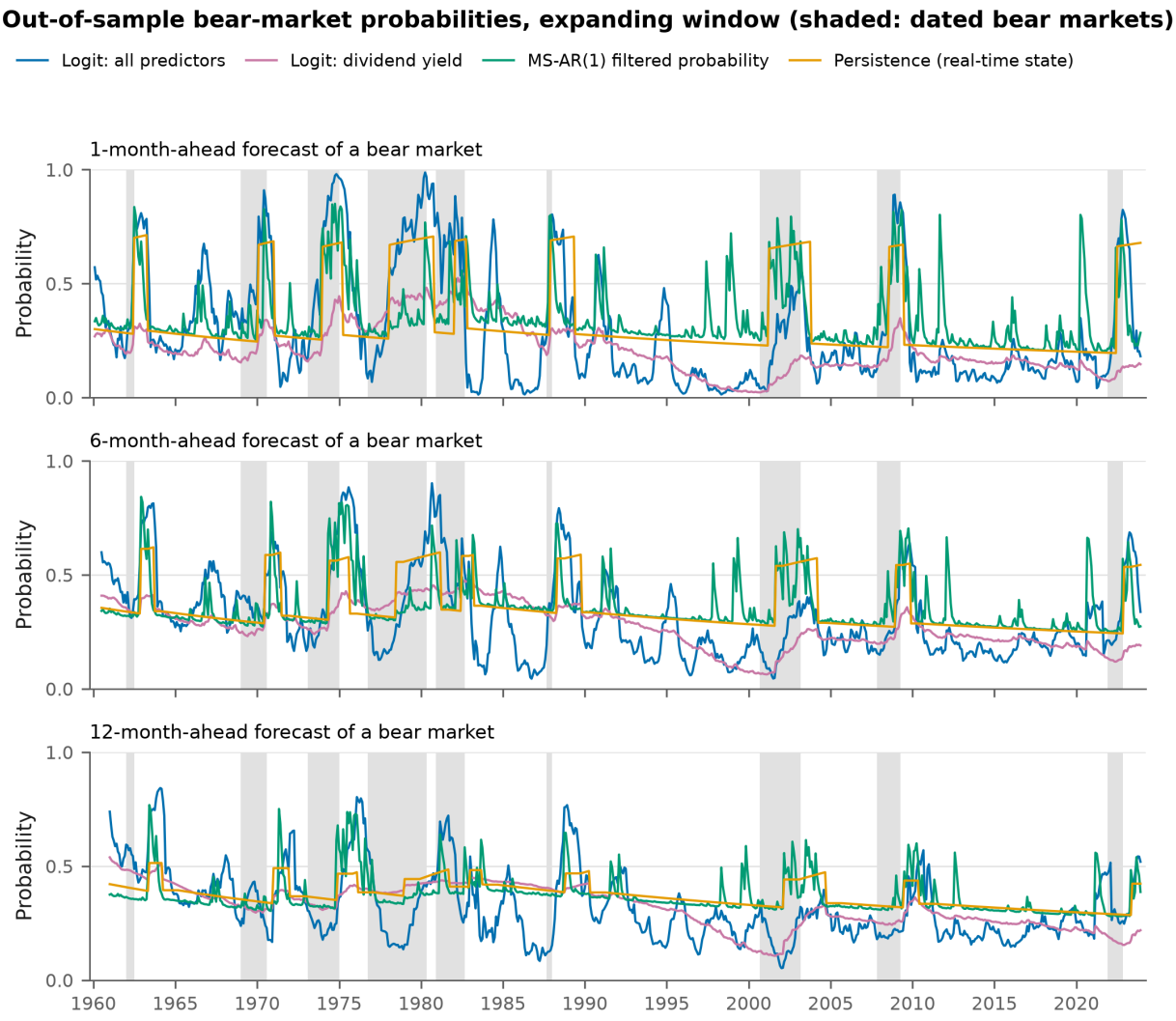


Figure 4: Real-time forecast probabilities, plotted at the target month, with dated bear markets shaded. Each forecast uses only data available at its origin.

full-sample regime parameters to construct the target, so it describes associations with the regime probability rather than tradeable forecasts.

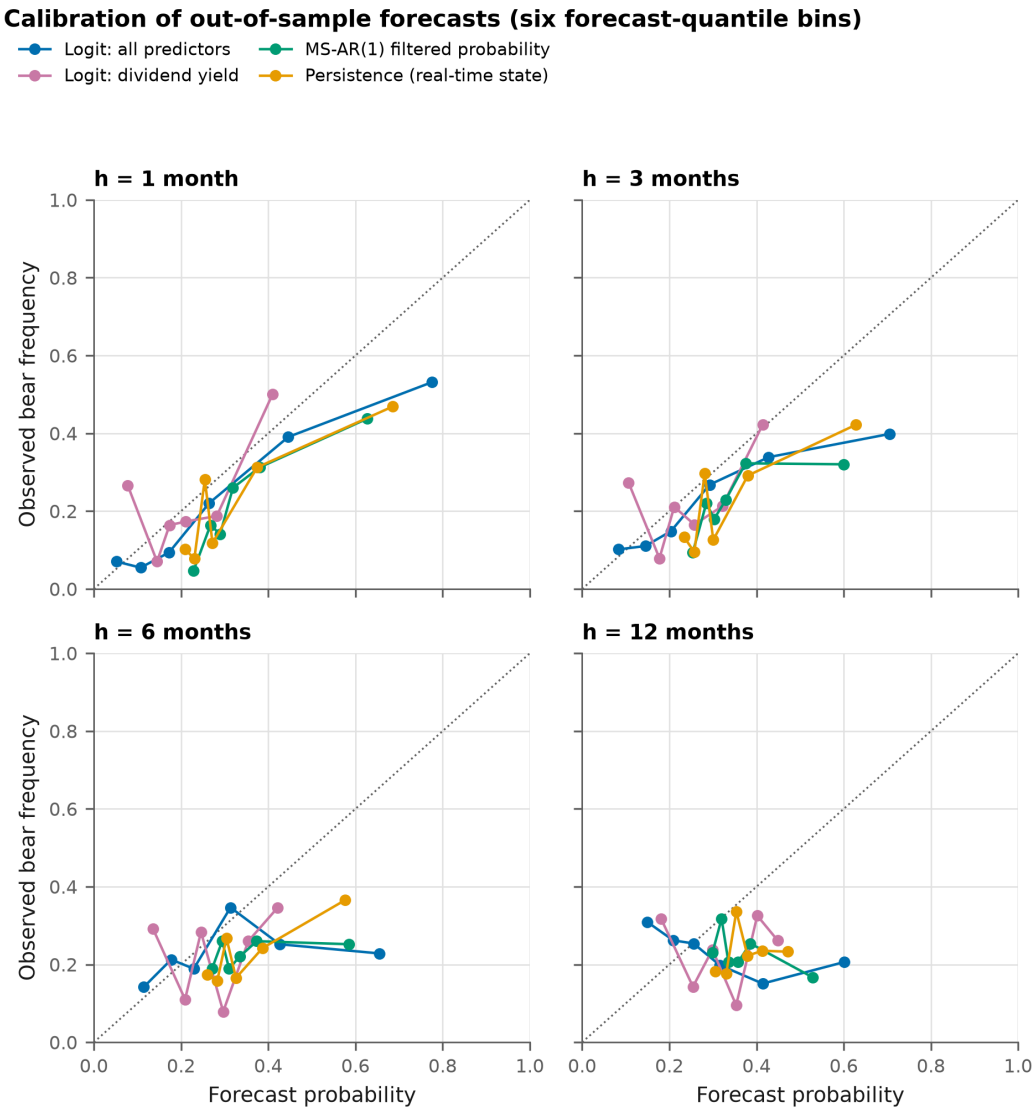
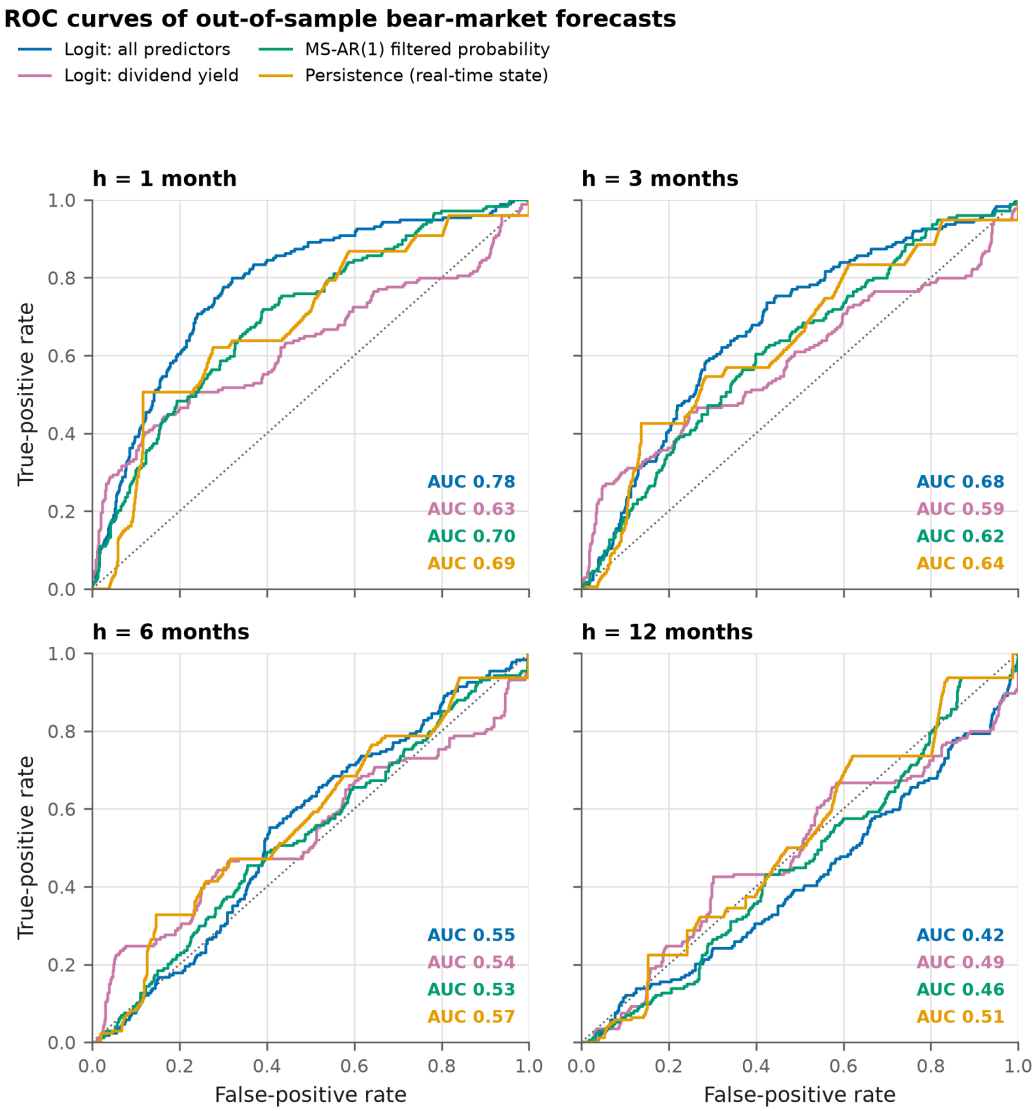


Figure 5: Calibration of out-of-sample forecasts: observed bear frequency against the mean forecast within six forecast-quantile bins. The dotted line is perfect calibration.



Source: Shiller, U.S. Stock Markets and CAPE Ratio data (archived extract one45.csv); author's calculations.

Figure 6: ROC curves of out-of-sample forecasts by horizon, with areas under the curve.

Table 7: Forecasting the regime probability: unrestricted versus intercept-only models

	OLS				Beta regression			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
A. Archived design and archived formula (squared mean error ratio)								
CPI	0.923	0.735	0.868	0.925	0.947	0.904	0.961	0.993
Long-term rate	0.981	0.975	1.008	0.997	0.992	0.981	1.002	0.996
Real bond	0.818	0.862	1.077	0.948	0.960	0.949	1.014	0.959
Real dividend	0.029	0.189	0.404	0.179	0.370	0.275	0.210	0.276
Real total return	0.006	0.921	0.942	0.982	0.564	1.019	0.956	1.021
TR-scaled earnings	0.363	0.474	0.759	1.365	0.819	0.837	0.894	1.222
TR-CAPE	0.822	1.016	1.018	1.027	0.706	1.010	1.004	1.013
B. Archived design, correct mean squared prediction error ratio								
CPI	1.002	1.021	1.005	1.006	0.996	0.996	0.997	1.000
Long-term rate	0.989	1.009	1.002	1.001	0.991	1.002	1.001	1.002
Real bond	0.997	1.041	1.019	1.011	0.970	1.027	1.006	1.015
Real dividend	1.154	1.184	1.148	0.963	0.951	0.924	0.883	0.806
Real total return	1.539	1.010	1.003	1.002	0.770	1.001	1.000	0.999
TR-scaled earnings	0.862	0.879	0.946	0.965	0.799	0.861	0.945	1.003
TR-CAPE	0.720	0.994	0.998	0.990	0.664	0.996	1.000	0.994
C. Corrected design: past-only predictors, true h-month lead, expanding window								
CPI inflation, monthly	1.003	0.951^{**}	0.918^{**}	0.939^{***}	0.976^{*}	0.930^{***}	0.920^{***}	0.935^{***}
Change in 10-year yield	1.002	1.000	0.985	1.005	1.004	0.998	0.990	1.003
Real bond return	1.007	1.003	0.987	1.012	1.008	0.997	0.999	1.010
Real dividend growth	0.953^{***}	0.965^{**}	0.935^{***}	0.941^{**}	0.934^{***}	0.943^{***}	0.930^{***}	0.935^{***}
Real total return	0.868^{***}	0.931^{***}	0.985^{*}	1.005	0.844^{***}	0.936^{***}	0.974^{**}	1.000
TR-scaled earnings growth	0.999	1.007	1.029	1.056	0.997	1.004	1.013	1.025
Change in log TR-CAPE	0.865^{**}	0.928^{***}	0.983^{**}	1.009	0.851^{***}	0.939^{***}	0.978^{**}	1.003
Benchmark: regime probability at t	0.476^{***}	0.872^{***}	1.032	1.025	0.483^{***}	0.853^{***}	0.985	1.004

Notes: Entries are ratios of the unrestricted model's loss to the intercept-only model's loss; values below one favor the predictor. Panels A and B re-run the archived code path (target Q(t-k), leaked regime model, single 80/20 split with the test period starting in 2011). Panel A uses the archived formula (mean error)² / (mean error)², which is not a mean squared error. Panel C forecasts Q(t+h) from predictors observable at t with expanding-window refits, origins 2011:04 to 2023:11 for h = 1. The benchmark row uses Q(t) itself as the predictor. Stars in Panel C mark Clark-West (2007) tests of equal accuracy in nested models where the unrestricted model is better: * p < 0.10, ** p < 0.05, *** p < 0.01.

7 The correction to the archived data alignment

The award version contains three errors, each verified by replication.

1. A future price in the regime model. The archived regime equation regresses $100 \ln P_t$ on the column `realpricelag`, which is $100 \ln P_{t+1}$. Every filtered probability from that model therefore conditions on next month's price.

2. Backward-looking "ahead" targets. The archived R script builds the "k months ahead" target with `lag(Q, k)` after loading `dplyr`, whose `lag` shifts a series back in time. The regressions therefore relate Q_{t-k}, the regime probability k months in the past, to the predictor's change at t. They describe how predictors move after the regime probability, not before it.

3. A mislabeled loss ratio. The archived "MSPE ratio" is $(\overline{e}_u)^2/(\overline{e}_r)^2$, the ratio of squared mean errors. Two models with small average bias can have a ratio near zero even when one has much larger squared errors.

Re-running the archived code path in Python, with all three errors, reproduces the award results (Figure 7, Table 8). The replicated transition probabilities are 0.950 and 0.833 against the reported 0.945 and 0.833 the correlation between replicated and reported values is 1.000 for the OLS coefficients, 1.000 for the R^2 statistics and 0.998 for the forecast ratios. The match confirms the diagnosis: the reported dividend coefficient at $k = 1$ is -0.195 and the replication with the backward target gives -0.198.

Table 8: Archived results, their replication, and the corrected estimates

	Archived (reported)	Replicated	Corrected
P(bull to bull)	0.945	0.950	0.944
P(bear to bear)	0.833	0.833	0.808
Dividend, k = 1: coefficient	−0.195	−0.198	−3.80*** (1.33) pp
Dividend, k = 6: coefficient	−0.325	−0.320	−3.40** (1.39) pp
Dividend, k = 6: R^2	0.075	0.074	0.017
Total return, k = 1: R^2	0.108	0.107	0.121
TR-scaled earnings, k = 1: R^2	0.085	0.085	0.001
Dividend, k = 1: OLS forecast ratio	0.004	0.029	0.953
Total return, k = 1: OLS forecast ratio	0.014	0.006	0.868
TR-scaled earnings, k = 1: OLS forecast ratio	0.390	0.363	0.999

Notes: Archived values are transcribed from the award PDF (v2/data/archived_reported.csv). 'Replicated' re-runs the archived code path in Python, including the future-price regressor, the backward target Q(t-k) and the archived MSPE formula. 'Corrected' uses the MS-AR(1) model on returns, past-only predictors, the forward target Q(t+k) and expanding-window forecasts. Coefficients in the corrected column are per one-standard-deviation change of a different (growth-rate) transformation and are therefore comparable in sign and significance only.

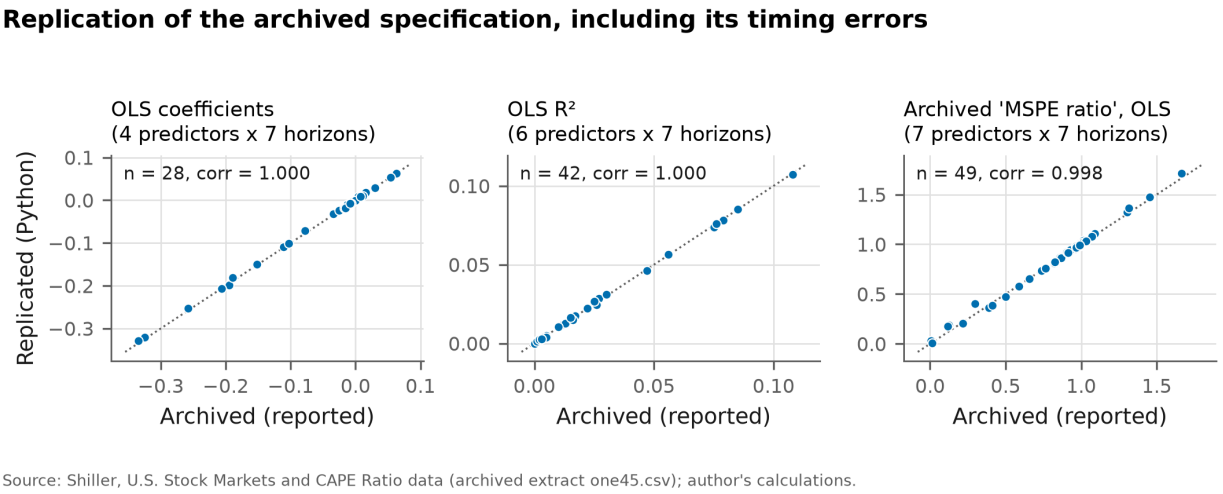


Figure 7: Replication of the archived specification. Each point compares a value reported in the award version with the same statistic from the Python replication of the archived code.

The effect of the corrections is largest for the headline out-of-sample claim. For real dividends at one month, the award version reported a ratio of 0.004, which the replication puts at 0.029. The true mean-squared-error ratio on the same design is 1.154, worse than the intercept-only model, and in the corrected design it is 0.953.

8 Limitations

The sample contains 9 bear markets since 1960, so the out-of-sample evaluation rests on few independent episodes and the overlapping monthly forecasts are strongly dependent; the Diebold-Mariano and Clark-West tests account for serial correlation only approximately. The bear-market rule is one reasonable choice among several, and monthly average prices understate intra-month declines: the 2020 decline, for example, does not reach the 20% threshold in monthly averages. Publication lags are imposed conservatively, but the Shiller series are themselves revised and partly interpolated, so this is a quasi-real-time rather than a true vintage exercise. The predictor set is limited to what the archived file contains; credit spreads and the term structure, which Chen (2009)

and Chen, Chen and Chou (2017) find useful, are not available in it. Finally, the regime-probability target in Table 4 and Table 7 uses full-sample parameters and should be read as descriptive.

9 Conclusion

Correcting the timing of the archived analysis changes its conclusions. The regime-switching description of the S&P 500 survives: real returns alternate between a long, calm bull regime and a short, volatile bear regime. The claim that market and macroeconomic indicators forecast bear markets does not survive in its original form. In real time, the information that best predicts a bear market next month is that one has already begun, and models built on inflation, valuation, momentum and volatility do not significantly improve on that persistence benchmark. Beyond a few months, the dividend yield retains a small and imprecise advantage over the unconditional frequency, and nothing else does. These results are consistent with a market in which downturns are difficult to anticipate from public information, and they illustrate how easily look-ahead in data alignment can create the appearance of predictability.

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